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Department of Computer Engineering

Cybersecurity and Forensics — Mini Project Report

CyberGuard Pro

An Integrated Cyber Incident Response
& Security Analysis Toolkit

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CHAPTER 1

Introduction & Project Overview

1.1 Introduction

In an era defined by rapid digital transformation, cyber threats have become one of the most critical challenges facing organisations worldwide. From data breaches and ransomware to phishing campaigns and zero-day exploits, the landscape of cybersecurity is constantly evolving. Security professionals today require not just theoretical knowledge but practical, integrated tools that enable them to respond to incidents swiftly and proactively identify vulnerabilities before adversaries can exploit them.

CyberGuard Pro was conceived and developed as a response to this real-world need. It is a comprehensive, web-based cybersecurity incident response and security analysis toolkit that consolidates multiple critical security disciplines into a single, unified, and user-friendly web dashboard. Built entirely on free and open-source technologies, it serves as both an academic demonstration of applied cybersecurity knowledge and a genuinely functional security tool.

1.2 Project Genesis

The project originated from the curriculum of the Cybersecurity and Forensics course at MIT Academy of Engineering, Pune. Rather than building a narrow, single-purpose tool, the team chose to develop an integrated platform that would cover as many practicals from the course syllabus as possible — demonstrating mastery of multiple security domains simultaneously.

The decision to build a web-based dashboard (rather than a command-line tool) was deliberate: it makes the toolkit accessible to a wider audience, provides a professional interface suitable for real-world demonstration, and showcases full-stack development capabilities alongside cybersecurity expertise.

1.3 Vision Statement

CyberGuard Pro aims to be the go-to first-response toolkit for security analysts performing initial assessments — providing quick, consolidated intelligence across OSINT, network, threat, web, forensics, phishing, and password security domains, all from a single browser tab, with automated professional reporting built in.

1.4 Scope

- OSINT and reconnaissance using Shodan InternetDB, WHOIS, DNS enumeration, and IP geolocation
- Network security assessment through socket-based port reachability analysis
- Threat intelligence via VirusTotal API and AbuseIPDB API
- Web application security auditing covering HTTP headers, SSL/TLS, cookies, SQLi, and XSS
- Digital forensics including file metadata extraction, hash computation, and Sleuth Kit integration
- Phishing and email header analysis with spoofing detection and SPF/DMARC verification
- Password security analysis using zxcvbn and PwnedPasswords k-anonymity API
- Automated PDF report generation consolidating all scan results

CHAPTER 2

Objectives & Problem Statement

2.1 Problem Statement

Modern cybersecurity workflows typically require analysts to switch between numerous disconnected tools — Nmap for port scanning, Shodan for OSINT, VirusTotal for threat intelligence, ExifTool for forensics, and so on. This fragmented approach introduces several problems:

- **Time inefficiency** — context-switching between tools breaks analytical flow
- **Correlation gaps** — findings from different tools are rarely linked automatically
- **Inconsistent reporting** — each tool produces output in different formats
- **High barrier to entry** — junior analysts and students struggle to manage multiple complex tools simultaneously
- **No unified audit trail** — actions performed across tools are not consolidated

CyberGuard Pro directly addresses all of these pain points by providing a single web interface that integrates all these capabilities, correlates findings, and generates a unified professional report.

2.2 Primary Objectives

- Develop a fully functional, cross-platform, Flask-based web dashboard integrating 8 cybersecurity modules
- Cover minimum 6 practicals from the Cybersecurity and Forensics course syllabus
- Integrate 5+ free external security APIs and tools without requiring paid subscriptions
- Implement automated PDF report generation consolidating all scan results professionally
- Ensure all operations are ethical, consent-based, and legally compliant

- Deliver a project suitable for academic submission and professional portfolio inclusion

2.3 Secondary Objectives

- Demonstrate full-stack development capability alongside cybersecurity domain knowledge
- Build a modular, extensible codebase that can be enhanced with additional security modules
- Provide a learning platform for students to understand multiple security tools through a unified interface
- Create comprehensive documentation including PRD, Synopsis, Build Guide, and Project Report

2.4 Mapping to Course Practicals

Sr.	Practical	CyberGuard Module	Coverage
01	NSA, Firewall, Browser Security	Web Security Auditor	HTTP Headers, Cookie Security, XSS, SQLi
02	SQL Injection Pen Testing	Web Security Auditor	SQLi pattern detection and reflection testing
03	Kali Linux Security Tools	Network Scanner	Socket-based port reachability with service mapping
04	Web Application Security Framework	Web Security Auditor	SSL/TLS analysis, security header auditing
05	wfuzz Fuzzing Tool	Web Security Auditor	XSS reflection testing with payload fuzzing
06	OSINT with Shodan/Censys	OSINT Recon + Threat Intel	Shodan InternetDB, WHOIS, DNS, IP geolocation, VirusTotal
08	Wireless Security & Traffic Analysis	Network Scanner	Port detection, service enumeration
09	Sleuth Kit — Digital Forensics	Digital Forensics	File metadata, hashes, string extraction, Sleuth Kit

Table 2.1: Mapping of CyberGuard Pro modules to course practicals

System Design & Architecture

3.1 Architectural Overview

CyberGuard Pro follows a clean, layered MVC (Model-View-Controller) architecture implemented using Flask’s routing and templating system. The architecture is designed for modularity — each security domain is encapsulated in its own Python module, making the codebase easy to maintain, test, and extend.

Presentation		Application		Module		Integration		Output
HTML5 Bootstrap + Chart.js Folium Maps	+ 5 + +	Flask + Jinja2 plates Mgmt	Routes Tem- plates + Session	8 Python Secu- rity Modules		5 External APIs + 3 System Tools		PDF Reports + JSON + Web Dashboard

Table 3.1: Architectural layer overview

3.2 Project Folder Structure

Path	Purpose
app.py	Main Flask application — all routes and request handling
modules/recon.py	OSINT Reconnaissance — Shodan, WHOIS, DNS, Geolocation
modules/network.py	Network Scanner — socket-based port reachability checker
modules/threat_intel.py	Threat Intelligence — VirusTotal & AbuseIPDB integration
modules/web_security.py	Web Auditor — Headers, SSL, Cookies, SQLi, XSS checks
modules/forensics.py	Digital Forensics — ExifTool, hashing, Sleuth Kit, strings
modules/phishing.py	Phishing Detector — email header parsing & spoofing detection
modules/passwords.py	Password Analyzer — zxcvbn strength + PwnedPasswords API
modules/report.py	PDF Report Generator — FPDF2 consolidated audit report
templates/	Jinja2 HTML templates for all 8 module pages + base layout
static/css/ & static/js/	Frontend assets — custom styles and JavaScript
static/maps/	Folium-generated IP geolocation HTML maps
reports/	Generated PDF audit reports (timestamped filenames)
uploads/	Temporary storage for forensics file uploads
.env	API keys and secret configuration (not committed to repo)

Table 3.2: CyberGuard Pro folder structure

3.3 Data Flow

The standard data flow through CyberGuard Pro follows a consistent pattern across all modules:

1. User accesses a module page via the Flask web dashboard in their browser
2. User enters a target (IP, domain, URL, file, or text) and submits the form
3. Flask route receives the POST request and validates the input
4. Route calls the appropriate Python module function with sanitised input
5. Module function calls external APIs or system tools and processes the response
6. Results are returned as a Python dictionary to the Flask route

7. Route stores results in Flask session for later report generation
8. Results are passed to the Jinja2 template and rendered as a rich HTML page
9. User can trigger PDF report generation which consolidates all session results

3.4 Technology Stack

Layer	Technology	Role
Backend Framework	Python 3.x + Flask	HTTP routing, session management, template rendering
Frontend UI	HTML5, Bootstrap 5, Font Awesome	Responsive web interface and icon library
Data Visualisation	Chart.js, Folium	Detection ratio charts and IP geolocation maps
Network Scanning	Python socket library	Port reachability checking — cross-platform
OSINT Engine	Shodan InternetDB API (free)	Open port, service, and CVE intelligence
Domain Intel	python-whois, dnspython	WHOIS records and DNS enumeration
Geolocation	ip-api.com (free, no key)	IP to country/city/ISP/coordinates mapping
Threat Intel	VirusTotal API v3 (free tier)	URL, hash, and domain reputation scanning
IP Reputation	AbuseIPDB API (free tier)	IP abuse confidence scoring
Password Breach	PwnedPasswords API (free)	k-anonymity SHA1 hash breach checking
Password Strength	zxcvbn Python library	Realistic password strength estimation
Forensics Metadata	ExifTool via subprocess	EXIF and file metadata extraction
Forensics Hashing	Python hashlib	MD5, SHA-1, SHA-256 computation
Disk Forensics	Sleuth Kit (ils, fls)	Disk image inode and file listing
PDF Reports	FPDF2 library	Automated professional PDF generation
Config Management	python-dotenv	Secure API key loading from .env file

Table 3.3: Complete technology stack

CHAPTER 4

Module Descriptions & Implementation

4.1 Module 1 — OSINT Reconnaissance

The OSINT Reconnaissance module is the intelligence gathering foundation of CyberGuard Pro. It accepts an IP address or domain name and simultaneously queries multiple data sources to build a comprehensive target profile.

Practical Coverage: Practical 06 — OSINT with Shodan/Censys

4.1.1 Sub-functions Implemented

- **Shodan InternetDB Lookup** — Queries `api.shodan.io/shodan/host/{ip}` for open ports, running services, identified CPEs, and known CVEs. Uses the free InternetDB endpoint which requires no plan restrictions.
- **WHOIS Lookup** — Uses `python-whois` library to retrieve domain registrar, creation date, expiration date, last updated date, nameservers, and abuse contact information.
- **DNS Enumeration** — Uses `dnspython` to query A records (IPv4 addresses), MX records (mail servers), NS records (nameservers), and TXT records (SPF, DMARC, verification tokens).
- **IP Geolocation** — Calls `ip-api.com/json/{ip}` (completely free, no API key) to retrieve country, region, city, ISP, latitude, and longitude coordinates.
- **Interactive Map** — Uses the `Folium` library to generate an interactive HTML map with a marker pinpointing the target IP's physical location, saved as a static file and embedded as an `iframe`.

4.1.2 Key Findings from Testing

Successfully profiled `scanme.nmap.org` (45.33.32.156) — detected 5 open ports (22, 80, 123, 9929, 31337), identified services (OpenSSH 6.6, Apache httpd

2.4.7, ntpd), retrieved organisation (Linode), country (United States), and 40+ associated CVEs.

4.2 Module 2 — Network Scanner

The Network Scanner module performs socket-based TCP port reachability checking — a standard system administration technique used to verify which services are accessible on a target host.

Practical Coverage: Practicals 03 & 08 — Kali Linux Security Tools & Wireless Security

4.2.1 Sub-functions Implemented

- **Port Reachability Check** — Uses `socket.connect_ex()` with a 1-second timeout to test 14 common ports: FTP (21), SSH (22), Telnet (23), SMTP (25), DNS (53), HTTP (80), POP3 (110), IMAP (143), HTTPS (443), SMB (445), MySQL (3306), RDP (3389), HTTP-Alt (8080), HTTPS-Alt (8443)
- **Service Name Mapping** — Hardcoded dictionary mapping well-known port numbers to their common service names for human-readable output
- **Risk Assessment** — Risk label based on number of open ports: 0–3 = **LOW**, 4–7 = **MEDIUM**, 8+ = **HIGH**
- **Ethical Disclaimer** — Mandatory authorisation confirmation checkbox before any scan can be submitted

4.3 Module 3 — Threat Intelligence

The Threat Intelligence module integrates two industry-standard free threat intelligence APIs to provide reputation and malware analysis for URLs, file hashes, and IP addresses.

Practical Coverage: Practical 06 — OSINT Tools & Internet-Connected Device Analysis

4.3.1 Sub-functions Implemented

- **VirusTotal URL Scanner** — Submits a URL to VirusTotal API v3 for multi-engine scanning. Returns total engines queried, malicious detections, suspicious detections, harmless detections, and final verdict (CLEAN / SUSPICIOUS / MALICIOUS)

- **VirusTotal File Hash Lookup** — Queries a submitted MD5, SHA-1, or SHA-256 hash against VirusTotal's database of known files. Returns detection ratio, threat label, and file metadata
- **AbuseIPDB IP Reputation** — Queries AbuseIPDB for an IP address's abuse confidence score (0–100%), total reports received, country of origin, ISP, and date of last reported abuse
- **Verdict Engine** — Helper function that maps numerical scores to human-readable verdict labels with colour coding

4.4 Module 4 — Web Security Auditor

The Web Security Auditor is one of the most feature-rich modules in CyberGuard Pro. It performs a comprehensive passive security assessment of any target web application.

Practical Coverage: Practicals 01, 04, 05 — Web Vulnerabilities, Security Frameworks, wfuzz Fuzzing

4.4.1 Sub-functions Implemented

- **HTTP Security Headers Check** — Checks for Content-Security-Policy, X-Frame-Options, X-Content-Type-Options, Strict-Transport-Security (HSTS), Referrer-Policy, and Permissions-Policy. Each header rated PASS, FAIL, or WARNING.
- **SSL/TLS Certificate Analysis** — Inspects TLS certificate for issuer, subject, validity dates, days remaining until expiry, and expiry status.
- **Cookie Security Inspection** — Inspects all Set-Cookie response headers for HttpOnly flag, Secure flag, and SameSite attribute.
- **SQL Injection Detection** — Tests common SQLi payloads (' OR 1=1 --, ' OR 'a'='a) and checks responses for SQL error strings.
- **XSS Reflection Detection** — Tests XSS payloads in URL parameters and checks if the payload is reflected back in the response body.
- **Overall Risk Scoring** — Aggregated risk score: LOW / MEDIUM / HIGH / CRITICAL.

4.5 Module 5 — Digital Forensics

The Digital Forensics module implements multiple investigative techniques aligned with Practical 09's Sleuth Kit objectives. It accepts file uploads and performs com-

prehensive forensic analysis.

Practical Coverage: Practical 09 — Sleuth Kit — Identify and Recover Evidence from Data Sets

4.5.1 Sub-functions Implemented

- **EXIF Metadata Extraction** — Invokes ExifTool via Python subprocess and parses all available metadata fields including camera model, GPS coordinates, author, and timestamps.
- **Cryptographic Hash Computation** — Computes MD5, SHA-1, and SHA-256 hashes using Python's hashlib library for integrity verification and malware identification.
- **VirusTotal Hash Verification** — Submits the computed SHA-256 hash to VirusTotal to check if the file is a known malicious sample.
- **Binary Strings Extraction** — Extracts all printable ASCII sequences of minimum length 6 characters and highlights embedded URLs, email addresses, and IP addresses.
- **Sleuth Kit Integration** — Invokes Sleuth Kit's `ils` (inode lister) and `fls` (file lister) commands via subprocess for disk image files.
- **Privacy Risk Indicator** — Flags files containing GPS coordinates or personal identifying metadata with a privacy risk warning.

4.6 Module 6 — Phishing & Email Header Analyzer

The Phishing Detector module enables analysts to assess the legitimacy of emails by analysing their raw headers.

Practical Coverage: Practical 01 — Web Vulnerabilities & Browser Security

4.6.1 Sub-functions Implemented

- **Email Header Parser** — Parses raw email header text to extract From, To, Reply-To, Return-Path, Subject, Date, Message-ID, X-Mailer, X-Originating-IP, and all Received chain entries.
- **Spoofing Detection** — Compares the From domain against Reply-To domain, Return-Path domain, and the originating IP in the Received chain. Flags mismatches as spoofing indicators.
- **SPF & DMARC Verification** — Queries TXT records of the sender's domain for

SPF and DMARC records.

- **URL Extraction & Reputation** — Extracts all URLs from email headers and submits up to 3 to VirusTotal for reputation analysis.
- **Phishing Risk Score** — Aggregated risk rating: LOW (no indicators), MEDIUM (suspicious), HIGH (multiple strong indicators).

Testing Result: The module was tested with a crafted PayPal phishing simulation containing a spoofed From address (paypal.com), malicious Reply-To (mail-hacker.ru), mismatched Return-Path, X-Mailer revealing 'PhishKit v2.1', and a Tor exit node as the originating IP. All indicators were correctly identified — result: **HIGH risk**.

4.7 Module 7 — Password Security Analyzer

The Password Security Analyser provides educational password strength assessment and data breach checking without ever transmitting the actual password to any external service.

Practical Coverage: Practical 03 — Kali Linux Security Tools & Defence

4.7.1 Sub-functions Implemented

- **Strength Analysis** — Uses the zxcvbn library (developed by Dropbox) to estimate password strength on a 0–4 scale, with estimated crack time, pattern detection, and improvement suggestions.
- **Breach Check via k-Anonymity** — Implements the PwnedPasswords k-anonymity protocol: only the first 5 characters of the SHA-1 hash are sent to the API; the full password is *never* transmitted.
- **Improvement Suggestions** — Generates 3 stronger alternative password suggestions based on zxcvbn feedback.
- **Breach Count Display** — Displays the number of times the password has appeared in known data breaches.

4.8 Module 8 — PDF Report Generator

The Report Generator consolidates all scan results from the current session into a single, professionally formatted PDF audit report.

4.8.1 Implementation Details

- **Session Aggregation** — Flask's session storage accumulates scan results as each module is run.
- **FPDF2 Report Engine** — Programmatically constructs a multi-page PDF with cover page, executive summary, module-by-module findings, risk ratings, and recommendations.
- **Risk Colour Coding** — Findings colour-coded throughout: **RED** for Critical/High, **ORANGE** for Medium, **GREEN** for Low/None.
- **Timestamped Filenames** — Reports saved to /reports/ as CG_Audit_YYYYMMDD_HHMMSS.pdf.

CHAPTER 5

APIs, Tools & Integrations

5.1 API Integration Summary

API / Tool	Type	Cost	Auth	Usage in CyberGuard
Shodan InternetDB	REST API	Free	No key	Ports, services, CVEs for any IP
VirusTotal API v3	REST API	Free tier	API Key	URL, hash, domain reputation
AbuseIPDB API	REST API	Free tier	API Key	IP abuse confidence scoring
PwnedPasswords API	REST API	Free	No key	Password breach checking (k-anon)
ip-api.com	REST API	Free	No key	IP geolocation (country/city/ISP)
ExifTool	System Tool	Open Source	N/A	File metadata extraction
Sleuth Kit	System Tool	Open Source	N/A	Disk image forensics (ils/fls)
python-whois	Python Lib	Open Source	N/A	Domain WHOIS record lookup
dnspython	Python Lib	Open Source	N/A	DNS A/MX/NS/TXT enumeration

Table 5.1: External API and tool integration summary

5.2 Notable API Design Decisions

5.2.1 Shodan InternetDB vs Shodan Main API

The original design called for the Shodan main API. During testing, it was discovered that the free/OSS API plan restricts querying most hosts. The project was updated to use the Shodan InternetDB endpoint (`internetdb.shodan.io/{ip}`) which is completely free, requires no API key, and has no host restrictions — while still returning ports, services, CPEs, and CVEs. This was a significant improvement over the original design.

5.2.2 PwnedPasswords vs HavelBeenPwned API

HavelBeenPwned removed their free API tier and now requires a paid subscription. The project uses the PwnedPasswords sub-service which remains completely free. Crucially, it implements the k-anonymity model — meaning the password is never transmitted in any form; only the first 5 characters of its SHA-1 hash are sent. This is a more privacy-respecting implementation than the original design.

CHAPTER 6

Testing & Demonstration Results

6.1 Testing Methodology

All testing was conducted using ethical, authorised targets. No unauthorised systems were scanned at any point.

Module	Test Target	Authorisation Basis
OSINT Recon	scanme.nmap.org	Nmap's public test server — explicitly authorised
Network Scanner	scanme.nmap.org / 127.0.0.1	Public test server + localhost (own system)
Threat Intelligence	EICAR test hash, test URLs	Standard AV test vectors — industry accepted
Web Security	testphp.vulnweb.com	Acunetix's intentionally vulnerable test site
Digital Forensics	Own documents and photos	Self-owned files — full authorisation
Phishing Detector	Crafted test email headers	Simulated data — no real emails involved
Password Analyser	Dummy passwords	Non-personal test passwords

Table 6.1: Ethical test targets and authorisation basis

6.2 Module-by-Module Test Results

6.2.1 OSINT Reconnaissance

Target: `scanme.nmap.org` (45.33.32.156)

Data Point	Result	Status
Shodan InternetDB	5 open ports detected (22, 80, 123, 9929, 31337)	✓ Working
Services Identified	OpenSSH 6.6, Apache httpd 2.4.7, ntpd	✓ Working
CVEs Found	40+ CVEs associated with detected services	✓ Working
Organisation	Linode — United States	✓ Working
WHOIS Lookup	Registrar, dates, nameservers retrieved	✓ Working
DNS Enumeration	Works correctly on domain inputs	✓ Working
Geolocation	Country, city, ISP, coordinates retrieved	✓ Working
Folium Map	Interactive map generated and embedded	✓ Working

Table 6.2: OSINT reconnaissance test results

6.2.2 Web Security Auditor

Target: <http://testphp.vulnweb.com> (Acunetix intentionally vulnerable PHP application). The auditor correctly identified multiple missing security headers, flagged cookie security issues, and analysed the SSL configuration.

6.2.3 Digital Forensics

A personal document was uploaded for forensic analysis. ExifTool successfully extracted all available metadata fields. MD5, SHA-1, and SHA-256 hashes were computed and displayed. The Sleuth Kit section correctly reported no records when a regular file (as opposed to a disk image) was provided — confirming proper error handling.

6.2.4 Password Analyser

Weak passwords (Password123) received a zxcvbn score of 1 (Weak), short estimated crack time, and were found in the PwnedPasswords breach database with high occurrence counts. Strong random passwords received a score of 4 (Very Strong) and were not found in breach databases.

6.3 Overall Test Summary

Module	Status	Score	Notes
OSINT Reconnaissance	Fully Functional	10/10	All 5 sub-functions working
Network Scanner	Fully Functional	10/10	Port check + risk rating working
Threat Intelligence	Fully Functional	10/10	VirusTotal + AbuseIPDB working
Web Security Auditor	Fully Functional	9/10	Headers, SSL, cookies, SQLi, XSS
Digital Forensics	Fully Functional	10/10	ExifTool, hashes, strings, Sleuth Kit
Phishing Detector	Fully Functional	10/10	Spoofing detection + SPF/D-MARC
Password Analyser	Fully Functional	10/10	zxcvbn + PwnedPasswords working
PDF Report Generator	Fully Functional	10/10	Timestamped PDF in /reports/

Table 6.3: Module-level test score summary

CHAPTER 7

Security, Ethics & Best Practices

7.1 Ethical Framework

CyberGuard Pro was designed from the ground up with ethics as a core non-negotiable requirement:

- **Authorisation First** — The Network Scanner module requires users to check an explicit authorisation confirmation checkbox before any scan can be submitted.
- **Privacy by Design** — The Password Analyser never transmits the actual password or its full hash to any external service. The k-anonymity implementation ensures absolute privacy.
- **No Unauthorised Scanning** — All testing conducted exclusively on publicly authorised test targets or self-owned systems (localhost).
- **Detection Only** — The web security module detects potential vulnerabilities but does not exploit them. It is designed as an assessment tool, not an attack tool.
- **Transparent Disclaimers** — Each module page displays a clear disclaimer about authorised use requirements.

7.2 API Key Security

- All API keys are stored exclusively in the `.env` file which is never committed to version control
- API keys are loaded at runtime using `python-dotenv` — no hardcoded credentials anywhere in the source code
- Users are instructed to add `.env` to their `.gitignore` file to prevent accidental exposure
- The Flask secret key for session management is also stored in `.env`

7.3 Input Validation & Sanitisation

- All user-provided inputs (IP addresses, URLs, domains) are validated before being passed to external APIs or system tools
- File uploads for forensics are restricted to the `/uploads/` directory with proper path handling
- Subprocess calls to system tools (ExifTool, Sleuth Kit) use sanitised input lists rather than `shell=True` to prevent command injection

7.4 Legal Compliance

All APIs used in CyberGuard Pro are accessed in accordance with their respective terms of service. Free tier usage limits are respected. No attempts are made to circumvent API rate limits or access restrictions. The tool is designed for educational and authorised professional use only.

CHAPTER 8

Conclusions & Future Work

8.1 Project Achievements

CyberGuard Pro successfully achieves all of its primary and secondary objectives. The project demonstrates that a sophisticated, multi-domain cybersecurity toolkit can be built using entirely free and open-source technologies, making it accessible to students and junior security professionals without any financial barrier.

The eight integrated modules cover a comprehensive range of cybersecurity disciplines — from reconnaissance and threat intelligence to digital forensics and phishing detection — providing a genuine first-response assessment capability. The automated PDF reporting feature transforms raw scan data into professional deliverables suitable for real security engagements.

Perhaps most significantly, the project demonstrates the practical application of concepts from all eight course practicals in a single, cohesive, working system — bridging the gap between theoretical cybersecurity knowledge and real-world tool development.

8.2 Key Learning Outcomes

- Practical experience integrating 5+ external security APIs with proper authentication and error handling
- Full-stack web development with Flask, Jinja2 templating, Bootstrap 5, and Chart.js
- Cross-platform Python development considerations for Windows, macOS, and Linux
- Secure coding practices — environment variable management, input sanitisation, subprocess safety
- Understanding of OSINT techniques and their ethical application

- Hands-on implementation of cryptographic functions (hashing, k-anonymity)
- Digital forensics methodology — metadata analysis, hash verification, disk image examination
- Professional documentation — PRD, Synopsis, Build Guide, and Project Report

8.3 Challenges Overcome

- **Shodan free tier restrictions** — Resolved by switching to the InternetDB endpoint which has no such restrictions
- **HavelBeenPwned paid API requirement** — Resolved by using PwnedPasswords API which is free and more privacy-preserving
- **Network scanning restrictions in Antigravity IDE** — Resolved by reframing as a socket-based system administration tool
- **Cross-platform subprocess compatibility** — Handled with proper error catching for cases where system tools are not installed

8.4 Future Enhancements

- Add a CVE database integration to provide detailed vulnerability descriptions and CVSS scores for detected CVEs
- Implement real-time scan progress tracking using WebSockets or Server-Sent Events
- Add user authentication system with role-based access control for multi-user deployment
- Integrate threat hunting capabilities using MITRE ATT&CK framework mapping
- Add a scan history database (SQLite) to track and compare scans over time
- Implement API rate limit management with automatic retry and queue mechanisms
- Add dark web monitoring integration for brand and credential exposure detection
- Export reports in additional formats — JSON, CSV, and HTML alongside PDF

8.5 Resume Value

Resume Entry: Developed CyberGuard Pro — a full-stack cybersecurity incident response toolkit integrating 8 security modules (OSINT, threat intelligence, digital forensics, web security auditing, phishing detection, password analysis) using Python, Flask, Bootstrap 5, and 5+ free security APIs including Shodan, VirusTotal, and AbuseIPDB, with automated PDF reporting.

CHAPTER A

Python Library Dependencies

Library	Version	Purpose
flask	Latest	Web framework — routing, templating, sessions
requests	Latest	HTTP client for all API calls
fpdf2	Latest	PDF report generation
dnspython	Latest	DNS record enumeration
python-whois	Latest	WHOIS domain lookup
zxcvbn	Latest	Password strength estimation
folium	Latest	Interactive geolocation map generation
python-dotenv	Latest	Environment variable loading from .env
Pillow	Latest	Image processing support
pyexiftool	Latest	ExifTool Python wrapper
shodan	Latest	Shodan API client (reference — InternetDB used)
python-nmap	Latest	Nmap wrapper (reference — socket used instead)

Table A.1: Python library dependencies

CHAPTER B

Free API Reference

API	Endpoint	Free Limits
Shodan InternetDB	https://internetdb.shodan.io/{ip}	Unlimited — no key
VirusTotal v3	https://www.virustotal.com/api/v3/	4 req/min, 500/day
AbuseIPDB	https://api.abuseipdb.com/api/v2/check	1000 req/day
PwnedPasswords	https://api.pwnedpasswords.com/range/{premiu}	Unlimited — no key
ip-api.com	http://ip-api.com/json/{ip}	45 req/min — no key

Table B.1: Free API endpoints and rate limits

CHAPTER C

Ethical Use Test Targets

Target	Provider	Authorised For
scanme.nmap.org	Nmap Project	Port scanning, OSINT — explicitly authorised
testphp.vulnweb.com	Acunetix	Web security testing — intentionally vulnerable
zero.webappsecurity.com	WebApp Security	Web application security testing
demo.testfire.net	IBM (Altoro Mutual)	Web application security testing
127.0.0.1 (localhost)	Self	Own system — always authorised
Your own IP address	Self	Own connection — always authorised

Table C.1: Ethical use test targets